

Monish Prasanna S

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PROFESSIONAL SUMMARY

B.Tech. Computer Science student (2027) specializing in **systems security** and **secure systems programming**. Hands-on experience building secure infrastructure (custom PKI/CA workflows, RFID authentication systems) and offensive security tools (USB payload delivery, HID emulation, protocol analysis). Active CTF competitor with focus on cryptography and reverse engineering. Seeking security engineering or application security internships where secure code and infrastructure defense are core priorities.

EDUCATION

Amrita Vishwa Vidyapeetham Chennai, India
B.Tech. in Computer Science and Engineering – Specialization in Cyber Security *Expected May 2027*
CGPA: 7.34/10 (First Class) | *Relevant Coursework: Network Security, Cloud Forensics, Cryptography, Compiler Design*

TECHNICAL SKILLS

Security Engineering: Custom CA/PKI Workflows, Protocol Analysis (Wireshark, websocat), HID Emulation, Physical Security Testing, Digital Forensics, Penetration Testing, Metasploit
Systems Programming: Rust (memory-safe cryptography, concurrent systems), C/C++, Embedded C (ESP32), PowerShell Scripting
Development & Tools: Python, SQL, Haskell, Docker, Git, Redis, Tailscale, Reverse SSH Tunneling
Hardware & IoT Security: ESP32 Microcontrollers, RFID/NFC Security (RC522), Hardware Attack Vectors, Network Protocol Analysis

SECURITY ENGINEERING PROJECTS

- USB Payload Delivery Framework** | *C, HID Protocol, PowerShell, Networking* 2025
 - Developed USB-based attack tool demonstrating credential exfiltration from locked Windows workstations using HID emulation and scripted keystroke injection.
 - Automated full attack chain (device insertion → credential extraction → remote exfiltration) with 100% success rate across Windows 10/11 environments in controlled testing.
 - Implemented anti-detection mechanisms including timing delays and AV-aware payload staging techniques.
- Custom Certificate Authority Infrastructure** | *OpenSSL, PKI, Cryptography* 2024 – Present
 - Designed and deployed custom CA infrastructure to research certificate validation attacks and trust chain vulnerabilities across browsers and operating systems.
 - Documented certificate pinning bypass techniques and explored MITM attack vectors in enterprise environments.
 - Built automated tooling for certificate generation, signing, and deployment across test infrastructure.
- ESP32 RFID Authentication System** | *Embedded C, IoT Security, Hardware* 2024
 - Built hardware-based access control system integrating RC522 RFID reader with ESP32 for secure audio system authentication.
 - Achieved <100ms authentication latency with zero false positives across 500+ test cycles using real-time tag verification.
 - Implemented comprehensive logging for unauthorized access attempts and anomaly detection for cloned tags.
- Memory-Safe Cryptographic Protocols** | *Rust, Cryptography* 2024
 - Implemented Schnorr signature scheme in Rust with focus on memory safety and resistance to timing side-channel attacks.
 - Designed constant-time operations and leveraged Rust's ownership model to prevent buffer overflows and use-after-free vulnerabilities.

DEVELOPMENT PROJECTS

- Chef Esoteric Language Compiler** | *Compiler Design, Lexical Analysis, Code Generation* 2024
- Designed 7-phase compiler implementing full compilation pipeline: lexical analysis, parsing, semantic checking, intermediate code generation, optimization, and code generation.
 - Built robust error handling with detailed diagnostics and recovery mechanisms for syntax and semantic errors.
- Secure URL Shortening Service** | *MERN Stack, Redis, Security* Jan 2026 – Mar 2026
- Architected production-grade URL shortener with MERN stack and Redis (Upstash) caching for sub-10ms redirect latency.
 - Hardened against abuse vectors: implemented rate limiting (100 req/hour/IP), SSRF protection, CSP headers, and malicious link detection; achieved 99.9% uptime handling 10,000+ redirects.

COMPETITIONS & RECOGNITION

- Top 5 Finalist – Finvest Pitching Competition** | *Shaastra 2025, IIT Madras* Jan 2025
- Selected among top 5 teams nationwide in India's largest student-run tech fest pitching competition.
 - Developed and presented financial technology solutions through multi-stage screening and live simulation rounds.
- CTF Competitor** | *Capture The Flag Security Competitions* 2024 – Present
- Active participant in CTF competitions focusing on cryptography, reverse engineering, binary exploitation, and web security challenges.

CERTIFICATIONS

- Palo Alto Networks Cybersecurity Foundation** | *Network, Cloud & Security Operations* Feb 2026
- Rust Systems Programming** | *Memory Safety & Concurrent Systems* Mar 2026